The Treasure Map Mystery

Whodunnit? · Transformations · Teacher Guide and Answer Key

Unit	Time	Supplies	Group size
Transformations	20–40 minutes	Pencil, scrap paper, scissors	Teams of 2–3 (or solo)

Skills practiced: Translations, reflections, rotations and dilations on the coordinate plane.

General procedures

1. Give each student the **student sheet** (story and case board) and the **Show Your Work** page.
2. Print the **Clue Cards** (pages 3–7 of the student packet) and cut them apart on the dashed lines. Laminating them lets you reuse the set. Post them around the room.
3. Students visit the cards in any order. At each card they solve the problem on their work page, find their answer among the four choices, read the clue beside it, and cross things off the case board. **Every wrong choice leads to a false clue** that clears the real culprit, so a mistake shows up when the board does not narrow down to one suspect, one place and one method.
4. When a team thinks it knows, they bring you the three names. If they are wrong, let them hunt for the card where something went wrong.

Solution: Explorer Rio took it, in the Rooftop Observatory, using this method: Sent as a paper airplane.

The reveal (read aloud): While members admired the empty case, **Explorer Rio** slipped onto the **Rooftop Observatory**, folded the Ancient Map into a **paper airplane**, and sailed it over the wall to a friend below. Every alibi and method you cleared led to Rio. X marks the spot!

Clue map: what each correct clue clears

Card	Clears
1	Suspect: Prof. Atlas · Place: Map Room
2	Method: Rolled in a poster tube
3	Suspect: Ms. Compass · Place: Rare Books Vault
4	Method: Folded inside a book
5	Suspect: Mr. Grid · Place: Globe Gallery
6	Method: Hidden in a globe
7	Suspect: Captain Nova · Place: Reading Nook
8	Method: Slipped in a backpack
9	Suspect: Nora Webb · Place: Courtyard Fountain
10	Method: Tucked in an umbrella

Answer Key

The Treasure Map Mystery · Correct choice and worked solution for each card

#	Problem	Correct choice	Worked solution	Clue it leads to
1	Translate the point (2, 3) using the rule (x, y) → (x + 4, y − 1). Find the image.	**C.** (6, 2)	(2 + 4, 3 − 1) (6, 2)	Prof. Atlas was drawing coastlines in the Map Room, with students watching.
2	Reflect the point (3, −2) across the x-axis. Find the image.	**C.** (3, 2)	reflecting across the x-axis changes the sign of y (3, 2)	Every poster tube was checked at the door. They held only posters.
3	Reflect the point (−4, 5) across the y-axis. Find the image.	**B.** (4, 5)	reflecting across the y-axis changes the sign of x (4, 5)	Ms. Compass was shelving ancient books in the Rare Books Vault. The vault log shows her key card.
4	Rotate the point (2, 3) 90° counterclockwise about the origin. Find the image.	**A.** (−3, 2)	rule: (x, y) → (−y, x) (−3, 2)	The books were all flipped through. No map folded inside.
5	Rotate the point (4, −1) 180° about the origin. Find the image.	**B.** (−4, 1)	rule: (x, y) → (−x, −y) (−4, 1)	Mr. Grid was measuring the Globe Gallery with a laser. The measuring log is time-stamped.
6	Dilate the point (3, −6) by a scale factor of $\frac{1}{3}$ with the center at the origin. Find the image.	**A.** (1, −2)	multiply both coordinates by $\frac{1}{3}$ (1, −2)	The big globe was opened. It held nothing but a dusty ruler.
7	Dilate the point (−2, 5) by a scale factor of 2 with the center at the origin. Find the image.	**D.** (−4, 10)	multiply both coordinates by 2 (−4, 10)	Captain Nova was reading in the Reading Nook. A guard checked on her twice.
8	Reflect the point (1, 4) across the line y = x. Find the image.	**B.** (4, 1)	reflecting across y = x swaps x and y (4, 1)	Every backpack was searched at the exit. No map inside.
9	The translation (x, y) → (x − 3, y + 2) moves point A to A′. If A = (5, 1), find A′.	**C.** (2, 3)	(5 − 3, 1 + 2) (2, 3)	Nora Webb was feeding the koi at the Courtyard Fountain with a tour group.
10	Rotate the point (−2, −5) 90° clockwise about the origin. Find the image.	**D.** (−5, 2)	rule: (x, y) → (y, −x) (−5, 2)	The umbrella stand holds umbrellas only, with nothing tucked inside.

Wrong-Answer Map

The Treasure Map Mystery · What a wrong choice looks like, and the false clue it leads to

Card	Wrong choice	Likely error	False clue
1	A. (4, −1)	used only the translation numbers	No paper airplane was found in the courtyard. The fountain stayed clear of paper.
1	B. (6, 4)	subtracted wrong sign on y	Explorer Rio was updating the world map in the Map Room. The curator saw his boots.
1	D. (−2, 4)	applied inverse translation	Prof. Atlas was tracing a compass rose on the Rooftop Observatory. A student held the lamp.
2	A. (−2, 3)	swapped coordinates	The paper airplane bin was empty. Nobody had folded one that day.
2	B. (−3, −2)	negated both	Explorer Rio was hunting for an old atlas in the Rare Books Vault. The vault log has his visit.
2	D. (−3, 2)	reflected over y-axis	Ms. Compass was reading star charts at the Rooftop Observatory. The telescope sign-in sheet shows her.
3	A. (5, −4)	swapped coordinates	Every window was shut and locked. No airplane could have been sent out.
3	C. (4, −5)	reflected over x-axis	Mr. Grid was surveying the Rooftop Observatory with a tripod. The tripod holes are still there.
3	D. (−4, −5)	negated both	Explorer Rio was napping in the Reading Nook with a travel book. A guard woke him.
4	B. (3, −2)	rotated clockwise	Explorer Rio was resting by the Courtyard Fountain and eating a sandwich. A pigeon agrees.
4	C. (−2, −3)	rotated 180	Captain Nova was checking the weather at the Rooftop Observatory. Her notes are dated.
4	D. (3, 2)	swapped coordinates	The rooftop breeze was too weak to carry any paper plane. Nothing was flown.
5	A. (−4, −1)	negated only y	Nora Webb was sweeping the Rooftop Observatory floor with a broom. The janitor confirms it.
5	C. (−1, −4)	rotated 90 clockwise	Not one paper airplane was found on the library grounds. Only leaves.
5	D. (4, 1)	reflected over x-axis only	Explorer Rio was unrolling maps in the Map Room all morning. The librarian counted them.
6	B. (1, 2)	sign error on y	A search of the lawn and roof found no folded paper. No plane was launched.
6	C. (9, −18)	multiplied by 3	Explorer Rio was spinning the globes in the Globe Gallery. A tour group watched him.
6	D. (−1, 2)	sign error	Prof. Atlas was pointing at the stars from the Rooftop Observatory with the visiting class.
7	A. (4, −10)	sign error on x	The paper airplane contest was canceled that day. Nobody launched a single plane.
7	B. (−4, −10)	sign error	Mr. Grid was lining up a compass reading at the Rooftop Observatory. The logbook has his initials.
7	C. (0, 7)	added scale factor	Explorer Rio was borrowing a pen in the Reading Nook. Ms. Compass lent him one.
8	A. (−4, −1)	reflected over y = −x	Explorer Rio was dusting a shelf in the Rare Books Vault. The key card swipe backs this up.

8	C. (−1, 4)	rotated 90 counterclockwise	The paper stack in the copy room was still sealed. No sheets were folded.
8	D. (1, −4)	rotated 90 clockwise	Mr. Grid was leveling the telescope at the Rooftop Observatory. The level reading is time-stamped.
9	A. (8, 3)	wrong sign on x	The gardener swept the courtyard clean and found no paper plane. Not one.
9	B. (2, −1)	wrong sign on y	Captain Nova was setting up a sky camera on the Rooftop Observatory. The camera clock shows it.
9	D. (8, −1)	used inverse translation	Explorer Rio was tossing coins at the Courtyard Fountain with a group of visitors.
10	A. (5, 2)	wrong sign	Ms. Compass was dusting the telescope in the Rooftop Observatory. Her cloth is still gray.
10	B. (5, −2)	rotated counterclockwise	Explorer Rio was tracing a trail on the big wall map in the Map Room. The guide took notes.
10	C. (−5, −2)	wrong sign on y	Every paper airplane in the lost-and-found was unfolded. All were plain homework.